Zihong (Logan) Luo

M.S. AI Engineering (ECE), Carnegie Mellon University | Available for Summer 2027 internships

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Focus: robot learning & simulation, machine learning systems, software engineering

Education

Carnegie Mellon University *Pittsburgh, PA, USA*

M.S. in AI Engineering – Electrical and Computer Engineering *Aug 2026 – Dec 2027 (Expected)*

- **Coursework:** 10-714 Deep Learning Systems (automatic differentiation, CUDA/GPU kernels, compiler-style computation graphs), 18-763 Systems & Tool Chains for AI Engineers, 18-661 Intro to ML for Engineers

University of Liverpool *Liverpool, UK*

BSc in Computing Science (Artificial Intelligence) | First Class Honours | GPA: 3.9/4.0 *Sep 2022 – Jun 2026*

Technical Skills

Languages: C++, Python, CUDA, C (embedded)
Simulation & Robotics: MuJoCo, Isaac Sim / Isaac Lab, LIBERO, LeRobot, ROS/ROS2, URDF/MJCF, FK/IK
Control: cascaded PID, IMU-based state feedback, balance control; real-robot teleoperation (Franka, UR5, SO-101)
ML Systems: PyTorch, Transformers, VLA/VLM training & evaluation, probing and mechanistic interpretability
Data Systems: PySpark, Kafka, Neo4j, NoSQL | **Hardware:** ESP32, Arduino, Raspberry Pi, IMU, LiDAR
Honors: Intl. Quant Championship, Top 0.1% (UK Finals 2025)

Research & Engineering Experience

Carnegie Mellon University *Pittsburgh, PA, USA*

Graduate Researcher – Lead Student Researcher (Advisor: Prof. Guannan Qu) *Sep 2026 – Present*

- Leading research on how pretrained Vision-Language Models (VLMs) encode spatial and physical knowledge, and whether these representations transfer to Vision-Language-Action (VLA) policies for robotic manipulation.
- Designing controlled experiments in LIBERO using Qwen3-VL and VLA models, with representation probing, counterfactual scene pairs, and activation interventions to investigate how geometric information influences downstream action generation.

SmartLab, University of Liverpool *Liverpool, UK*

Research Assistant (Advisor: Prof. Anh Nguyen) *Oct 2025 – Aug 2026*

- Core contributor to **GVLA** (**ECCV 2026**, 2nd author): a VLA that conditions manipulation on **gripper morphology**; co-designed the gripper-conditioning module from inception, implemented and ablated multiple architectural variants, ran experiments, and co-wrote the paper.
- Worked across **simulation and real hardware** on multi-gripper Franka / UR5 platforms with dual RGB-D cameras: teleoperation, data collection, behavior cloning, and sim-to-real evaluation.
- **GCA-Bench** (**IROS 2026**): complex-grasping benchmark from detection to execution; contributed real-robot data collection, dataset organization, and manuscript preparation.

Donkey (Y Combinator S26) [YC] *Remote*

Software Engineering Contributor *Sep 2025 – Jan 2026*

- Contributed to full-stack development (frontend, backend, databases) and AI-agent orchestration for an AI-native sourcing platform. `[product]`

Jifu Medical (AI Algorithm Group) *Shenzhen, China*

Algorithm Intern – Robotics & Multimodal Perception *Jun 2025 – Aug 2025*

- Built an end-to-end imitation-learning pipeline (**ACT / ALOHA / LeRobot**) for **SO-101 dual-arm** manipulation: teleoperation, synchronized multi-view recording, and joint-state/action logging; collected 50+ episodes.

MBZUAI (Remote) **&** **University of Exeter** (Remote) *Research Assistant*

Advisors: Prof. Imran Razzak; Prof. Yanda Meng *Mar 2024 – Jul 2025*

- **PG-SAM** (**BIBM 2025**, co-first author): designed a Modality Prior Aligner and fusion decoder linking LLM-derived semantic priors to pixel-level segmentation.
- **IMDR** (**AAAI 2025 Oral**): designed ablations and incomplete-modality evaluations for multimodal representation learning.

Simulation & Control Projects

Single & Double Inverted Pendulum: Dynamics Simulator & Control | *JavaScript* | `[code]` *2026*

- Implemented single-/double-pendulum dynamics with 1 kHz **RK4** integration, friction, actuator saturation, and contact handling.
- Developed energy-shaping swing-up and pole-placement balance controllers with supervisory switching.

Self-Balancing Wheeled-Legged Robot | *C++, ESP32, Arduino* *2025*

- Built end-to-end: leg inverse kinematics, IMU attitude feedback, and a **cascaded PID** balance controller in embedded C++.

Selected Publications

Robotics: Gripper-aware VLA (**ECCV'26**, 2nd author) · GCA-Bench (**IROS'26**) · Graph-Guided Gripper-Aware VLA (**ICRA'27**, under review)
Other: MC-DBN (**ICPR'24**, co-first) · PG-SAM (**BIBM'25**, co-first) · IMDR (**AAAI'25 Oral**) · +2 workshop/journal [full list]